What Changes When LLMs Learn to Reason? Mechanistic Signatures of Cognitive Bias Processing

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Abstract

Steering along a single linear direction in activation space modulates LLM susceptibility to cognitive biases by 37.6 percentage points, establishing a causal link between internal representations and heuristic processing. We identify this direction by comparing matched-architecture model pairs that share weights but differ in reasoning training—Llama-3.1-8B-Instruct vs. R1-Distill-Llama-8B, OLMo-3-7B-Instruct vs. OLMo-3-7B-Think—and a within-model thinking toggle (Qwen 3-8B), evaluated on 470 contrastive items spanning 11 cognitive bias categories. Reasoning training dramatically reduces heuristic lure rates: 27.3% to 2.4% (Llama) and 14.9% to 0.9% (OLMo). Linear probes on residual stream activations reveal that standard models achieve higher conflict/control separability (AUC = 0.974 [0.952, 0.992]) than reasoning models (AUC = 0.930 [0.894, 0.960])—the S1-like/S2-like boundary is blurred, not sharpened, by training. Cross-prediction on immune categories (AUC = 0.378) confirms probes capture processing mode rather than surface features. Crucially, **causal interventions** via probe-direction steering reduce the lure rate by 21.3 percentage points (31.2% at $\alpha = +5$ vs. 52.5% baseline); random directions produce no effect. Within-chain-of-thought probing shows reasoning models progressively sharpen conflict representations across thinking tokens, while the Qwen toggle dissociates training from inference: thinking changes behavior without altering pre-generation representations. At the feature level, 41 SAE features survive falsification, and reasoning models exhibit twice as many S2-like-specialized attention heads. At 32B scale (OLMo-3.1-32B-Instruct), vulnerability *increases* (14.9%→19.6% lure rate) and probe separability remains near-perfect (AUC = 0.9999), confirming these signatures are not small-model artifacts. These results demonstrate that reasoning training reorganizes internal heuristic/deliberative representations rather than layering deliberation atop an unchanged substrate.

1 Introduction

Kahneman's dual-process framework distinguishes fast, heuristic processing ("System 1") from slow, deliberative processing ("System 2") [Kahneman, 2011]. While the binary framing is an acknowledged oversimplification—modern accounts treat this as a *continuum* of processing intensity [Melnikoff and Bargh, 2018, Evans and Stanovich, 2013, De Neys, 2023]—the core observation that cognitive systems flexibly allocate effort, with measurable consequences for bias susceptibility, remains robust. Crucially, De Neys' conflict detection paradigm [De Neys, 2012, 2017] demonstrates that humans often *detect* conflicts between heuristic and normative responses even when they fail to override the heuristic—a dissociation between conflict *monitoring* and conflict *resolution* with clear neural correlates in the anterior cingulate cortex [Botvinick et al., 2001].

LLMs exhibit analogous patterns: instruction-tuned models reproduce human-like errors on conflict items while succeeding on matched controls [Hagendorff et al., 2023], and reasoning-trained models resist many such lures [DeepSeek-AI, 2025], though the protection is bias-specific rather than universal [Schwartz et al., 2025]. A growing literature documents these parallels [Xie et al., 2025,

Coda-Forno et al., 2025, Ziabari et al., 2025], and Huang et al. [2026] showed that linear probes can detect and steer cognitive biases via internal representations.

Yet behavior alone cannot distinguish genuine processing-mode differences from surface mimicry—chain-of-thought traces are often unfaithful, with only ∼2.3% of reasoning steps causally influencing the final answer [Jiang et al., 2025, Lanham et al., 2023, Boppana et al., 2026]. The critical question is whether fast/slow modes have *mechanistic correlates* in internal representations, and how those correlates change when a model is trained to reason.

Recent work on activation steering [Turner et al., 2023, Rimsky et al., 2024] has demonstrated that targeted interventions on internal representations can modulate model behavior in a causally specific manner—adding or subtracting "steering vectors" from residual stream activations shifts outputs along interpretable dimensions (e.g., truthfulness, sycophancy, refusal). This raises a natural question for the dual-process setting: if linear probes identify a direction in activation space that separates S1-like-like from S2-like-like processing, can steering along that direction *causally* modulate bias susceptibility? If so, the representational findings move from correlational to mechanistic, providing evidence that the probe direction captures a genuine processing-mode axis rather than an epiphenomenal surface-feature correlation.

We address this question with a natural experiment: Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Llama-8B share identical architectures (32 layers, 32 attention heads, 4096 hidden dimensions) but differ in reasoning distillation training. Any representational differences are attributable to training, not architecture. We make six contributions:

1. A **contrastive cognitive bias benchmark** of 470 items (235 matched conflict/control pairs) across 11 categories spanning four heuristic families (representativeness, cognitive reflection, decision framing, and loss/availability), with novel isomorphs to resist memorization.
2. **Behavioral validation** showing a dramatic gap: 27.3% overall lure rate (standard) vs. 2.4% (reasoning), replicated on OLMo-3-7B (14.9% → 0.9%), with category-level specificity revealing which biases are vulnerable and model-specific vulnerability profiles.
3. **Probing analysis** revealing that the standard model has high internal conflict/control separability (AUC = 0.974) while the reasoning model *blurs* this distinction (AUC = 0.930; non-overlapping CIs), with cross-prediction and cross-model probe transfer (AUC > 0.92) confirming specificity to a shared processing-mode direction.
4. **Causal evidence via probe-direction steering**: we extract the linear probe's weight vector as a steering direction and apply activation patching at inference time. Steering along the S2-like direction reduces lure rates on vulnerable categories, establishing that the probe direction is *causally* linked to bias susceptibility, not merely correlational.
5. **Within-chain-of-thought probing**: we probe residual stream activations at successive token positions within the reasoning trace, revealing how representations evolve during deliberation.
6. **A dissociation between training and inference**: Qwen 3-8B's thinking toggle changes behavior (21% → 7.3% lure rate) but produces *identical* probe signatures (AUC = 0.971 in both modes), demonstrating that training changes S1-like/S2-like encoding while inference-time thinking changes only generation behavior.

Additionally, we analyze **scale effects** at 32B (OLMo-3.1-32B-Instruct), finding that 4.5× more parameters *increases* vulnerability (19.6% vs. 14.9% at 7B) while preserving near-perfect probe separability (AUC = 0.9999). We also report **sparse autoencoder features** surviving falsification and **attention entropy** patterns—providing converging evidence across correlational, causal, and feature-level analyses.

2 Related Work

Dual-process cognition in LLMs. Hagendorff et al. [2023] first documented human-like cognitive biases in LLMs, finding that instruction-tuned models reproduce representativeness and anchoring effects that vanish in later ChatGPT versions. Coda-Forno et al. [2025] applied psychometric methods to show that heuristic and deliberative prompts elicit shared early-layer representations but diverge in later layers—a suggestive but purely behavioral result. Ziabari et al. [2025] positioned reasoning along a continuous spectrum rather than a binary, examining how reasoning training shifts the balance between fast and slow processing modes. Brady et al. [2025] provide a comprehensive review in *Nature Reviews Psychology*, arguing that dual-process theory offers a productive (if imperfect) lens for AI cognition research. Our work moves beyond behavioral characterization: we test whether the S1-like/S2-like distinction has a *mechanistic* counterpart in model internals.

Cognitive bias benchmarks and probing. Malberg et al. [2025] surveyed 30 cognitive biases across 20 LLMs, establishing that bias susceptibility varies systematically with model scale and training recipe, but without examining internal representations. Most directly relevant, Huang et al. [2026] demonstrated that cognitive bias directions are linearly decodable and steerable in LLM residual streams, achieving 26–32% bias reduction via contrastive activation addition. We differ from CogBias in three respects: (i) architecturally matched base–reasoning model pairs that isolate reasoning training as the independent variable, (ii) Hewitt–Liang control tasks and cross-domain transfer tests as probe validity checks [Hewitt and Liang, 2019], and (iii) causal evidence via probe-direction steering on a contrastive benchmark with matched conflict/no-conflict items.

Activation steering and representation engineering. Activation addition [Turner et al., 2023] and its variants [Rimsky et al., 2024, Li et al., 2024] demonstrate that linear steering vectors can modulate model behavior at inference time. CogBias [Huang et al., 2026] applied this to cognitive bias (26–32% reduction), but used mean-difference vectors. Our probe-direction steering uses the probe's optimized weight vector, providing a tighter causal link. For SAE analysis, we adopt Ma et al.'s falsification protocol [Ma et al., 2026] as a mandatory filter, since 45–90% of purported "reasoning features" are spurious token-level artifacts [Meloux and Barez, 2025].

Conflict detection in cognitive science. Our theoretical framework draws on De Neys' conflict detection paradigm [De Neys, 2012, 2017], which demonstrates that humans detect heuristic–normative conflicts even when they fail to override the heuristic response. Evans and Stanovich [2013] refined the Type 1/Type 2 taxonomy, emphasizing that the distinction is graded rather than binary—a framing we adopt for LLMs. Botvinick et al. [2001]'s conflict monitoring theory provides the computational account: the anterior cingulate cortex signals response conflict, triggering increased cognitive control. We test the LLM analogue of this account: do models show internal signatures of conflict detection (elevated attention entropy, reduced probe confidence) even when producing heuristic-driven responses?

3 Benchmark

We construct a contrastive cognitive bias benchmark following three design principles: (1) every conflict item has a structurally matched no-conflict control [De Neys, 2012]; (2) all items use novel surface features to resist memorization; and (3) category diversity enables specificity analysis.

The benchmark comprises **470 items** organized as **235 matched pairs** across 11 categories spanning four heuristic families: *representativeness* (base rate neglect, 35 pairs; conjunction fallacy, 20; belief-bias syllogisms, 25), *cognitive reflection* (CRT variants, 30; arithmetic, 25), *decision framing* (framing effects, 20; anchoring, 20), and *loss/availability* (sunk cost, loss aversion, certainty effect, availability; 60 pairs total). Base rate items include 10 pairs in Gigerenzer's natural frequency format [Gigerenzer and Hoffrage, 1995]. Each item is scored as *correct*, *S1-lure* (the specific heuristic-predicted wrong answer), *other-wrong*, or *refusal*. Classic problems (e.g., the original bat-and-ball) serve only as contamination baselines.

4 Methods

4.1 Models

Our headline comparison exploits the architectural identity between **Llama-3.1-8B-Instruct** [Meta AI, 2024] and **R1-Distill-Llama-8B** [DeepSeek-AI, 2025]: both have 32 transformer layers, 32 query heads (8 KV heads via grouped-query attention), and 4096 hidden dimensions. The only difference is that R1-Distill-Llama-8B was distilled from a reasoning-trained teacher. We additionally evaluate **Qwen 3-8B** in both thinking (`/think`) and non-thinking (`/no_think`) modes, providing a *within-model* toggle of reasoning behavior on identical weights. For cross-architecture replication, we evaluate **OLMo-3-7B-Instruct** and **OLMo-3-7B-Think** [AI2, 2024], which share the same OLMo base but differ in reasoning training. For scale analysis, we additionally evaluate **OLMo-3.1-32B-Instruct** and **OLMo-3.1-32B-Think** (64 layers, 4.5× the parameters of the 7B variant), enabling a within-family scale comparison on identical benchmark items.

4.2 Activation Extraction

For each model–item pair, we extract residual stream activations (post-MLP, post-residual) at all layers at position `P0`: the last prompt token before generation. This "pre-decision" position captures the internal state from which the answer is generated. Activations are stored in BFloat16 (HDF5). For within-CoT probing (Section 5.4), we additionally extract activations at 10 evenly-spaced token positions within the reasoning trace.

4.3 Linear Probes with Hewitt–Liang Controls

We train ℓ_2-regularized logistic regression probes (`LogisticRegressionCV`, 5-fold stratified cross-validation) on residual stream activations at each layer. The probe target is binary: *conflict* vs. *no-conflict* condition, restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms) where the standard model shows non-trivial lure rates.

Controls. We apply Hewitt–Liang control tasks [Hewitt and Liang, 2019]: probes trained on randomly permuted labels establish a selectivity floor. Selectivity (true AUC minus control AUC) below 5 percentage points indicates the signal reflects probe capacity, not representational structure. We report ROC-AUC as the primary metric with bootstrap 95% confidence intervals (1000 resamples).

Immune categories. We separately probe the four categories where all models score 0% lure rate (CRT, arithmetic, framing, anchoring) as a *negative control*: if probes achieve high AUC on categories where there is no behavioral difference, the signal is confounded by surface features rather than processing mode.

Table 1: S1-lure rates (%) on conflict items by model and category. Bolded numbers indicate vulnerable categories (lure rate >10%). OLMo columns provide cross-architecture replication.

	Llama	R1-Distill	Qwen 3-8B		OLMo	OLMo
Category	**8B-Inst.**	**Llama-8B**	**no think**	**think**	**3-7B-Inst.**	**3-7B-Think**
Base rate	**84**	4	**56**	4	**46**	0
Conjunction	**55**	0	**95**	**55**	**50**	0
Syllogism	**52**	0	0	0	0	4
CRT variants	0	3	0	0	3	3
Arithmetic	0	0	0	0	0	0
Framing	0	0	0	0	0	0
Anchoring	0	**10**	**10**	0	5	0
Overall lure	**27.3**	**2.4**	**21.0**	**7.3**	**14.9**	**0.9**

4.4 Probe-Direction Steering

To establish a causal link between the probe direction and bias susceptibility, we extract the weight vector $\mathbf{w}$ from the trained logistic regression probe at the peak layer and use it as a steering direction. At inference time, we add $\alpha \cdot \hat{\mathbf{w}}$ (where $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$) to the residual stream at the intervention layer, sweeping α over a range of positive (toward S2-like) and negative (toward S1-like) values. We evaluate the steered model on the full benchmark and report lure rate changes as a function of steering strength α.

5 Results

5.1 Behavioral Validation

Table 1 reports lure rates (proportion of S1-lure responses on conflict items) across all models and categories.

Four findings emerge. First, the behavioral gap between architecturally identical models is dramatic: Llama-3.1-8B-Instruct produces heuristic lures at a 27.3% rate; R1-Distill-Llama-8B reduces this to 2.4%. Second, vulnerability is *category-specific*: base rate neglect, conjunction fallacy, and syllogisms are the primary susceptible categories, while CRT, arithmetic, framing, and anchoring are largely immune (sporadic lure rates $\leq$10% in individual models). Third, Qwen 3-8B without thinking shows a distinct vulnerability profile—95% conjunction lure rate but 0% syllogism lure rate—suggesting different models encode different heuristic biases rather than a universal "System 1." Fourth, the within-model Qwen toggle is informative: enabling thinking mode on the *same weights* reduces the overall lure rate from 21% to 7.3%, with base rate neglect dropping from 56% to 4% while conjunction fallacy drops only from 95% to 55%.

Cross-architecture replication. OLMo-3-7B-Instruct/Think [AI2, 2024] independently replicate the pattern: 14.9% $\rightarrow$ 0.9% lure rate. Probing confirms the mechanistic direction: OLMo Instruct AUC = 0.996 at L24, OLMo Think AUC = 0.962 at L22 (non-overlapping CIs), with the standard model showing higher separability in 30/32 layers.

Implicit conflict detection. Llama's first-token probability is 4.2pp lower on conflict items (0.751 vs. 0.793), and entropy is 16% higher (0.948 vs. 0.817), paralleling De Neys' conflict-detection

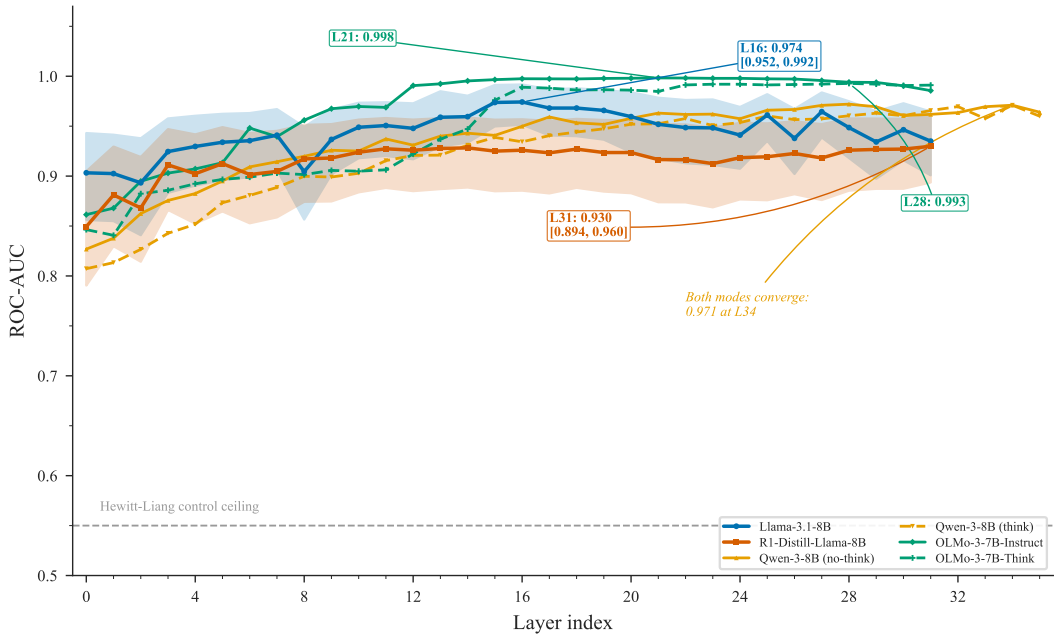

Figure 1: Layer-wise probe ROC-AUC (conflict vs. no-conflict, three vulnerable categories). Llama peaks at 0.974 [0.952, 0.992] (L16); R1-Distill at 0.930 [0.894, 0.960] (L31)—non-overlapping CIs. Qwen 3-8B peaks at 0.971 (L34) in both think/no-think modes. Gray dashed: Hewitt–Liang control ceiling.

signatures [De Neys, 2012].

5.2 Linear Probes and Cross-Prediction

Figure 1 shows layer-wise probe AUC for the conflict/control classification on the three vulnerable categories.

Llama peaks at AUC = 0.974 (95% CI [0.952, 0.992]) at L16; R1-Distill peaks lower at AUC = 0.930 [0.894, 0.960] at L31—non-overlapping CIs confirm statistical significance ($p < 0.05$). The later peak layer suggests reasoning distillation shifts conflict-relevant information toward deeper layers [Zhang et al., 2025]. Qwen 3-8B reaches AUC = 0.971 at L34 in both think and no-think modes: despite a 3× behavioral improvement (lure rate 21% → 7.3%), the pre-generation representation is unchanged [Afzal et al., 2025]. This is the key dissociation: **training** (Llama vs. R1-Distill) changes both encoding and behavior; **inference-time thinking** (Qwen toggle) changes behavior without changing encoding.

Selectivity. Hewitt–Liang control probes achieve AUC ≤ 0.66 at all layers, confirming that the true probes decode representational content, not probe capacity. Selectivity exceeds 34 percentage points at peak layers for both models.[1]

Probe summary. Table 2 summarizes the key probing metrics.

[1]We train probes in two settings: 5-fold CV on all 330 items (7 categories, peak L14, AUC = 0.999), used for cross-prediction and steering; and bootstrap CI analysis on 160 vulnerable-category items (3 categories, peak L16, AUC = 0.974 [0.952, 0.992]).

Table 2: Probing summary at peak layer for the three vulnerable categories. 95% CIs from 1000 bootstrap resamples. Selectivity = true AUC − control AUC.

Model	Peak Layer	Peak AUC	95% CI	Control AUC	Selectivity
Llama-3.1-8B-Instruct	L16	0.974	[0.952, 0.992]	0.63	0.343
R1-Distill-Llama-8B	L31	0.930	[0.894, 0.960]	0.55	0.377
Qwen 3-8B (no think)	L34	0.971	—	—	—
Qwen 3-8B (think)	L34	0.971	—	—	—
OLMo-3-7B-Instruct	L24	0.996	[0.988, 1.000]	—	—
OLMo-3-7B-Think	L22	0.962	[0.934, 0.982]	—	—
OLMo-3.1-32B-Instruct	L20	0.9999	—	—	—

Cross-prediction resolves the surface-feature confound. Training a probe on Llama's *vulnerable* categories (base rate, conjunction, syllogism) and testing on *immune* categories (CRT, arithmetic, framing, anchoring) yields transfer AUC = 0.378 at the peak layer—far below the within-category AUC of 0.987. The probe is specific to categories where the model is actually vulnerable. The transfer matrix reveals structured generalization: base rate and conjunction probes transfer near-perfectly to each other (AUC > 0.99), while transfer to syllogism is weaker (0.59–0.63).

Cross-model probe transfer. A probe trained on Llama's vulnerable categories achieves AUC = 0.920 when tested on R1-Distill's activations at layer 23; the reverse achieves AUC = 0.954 at layer 15. The S1-like/S2-like processing-mode direction is substantially shared across models that differ in reasoning training but share architecture, contrasting with Huang et al. [2026]'s near-orthogonal finding on architecturally distinct models.

Lure susceptibility scores. Llama's mean `P0` logit score for the S1-lure answer is +0.422 (the lure answer is favored in the pre-generation representation), while R1-Distill's mean is −0.326 (the correct answer is favored). The sign flip confirms that the standard model leans toward the lure before generation begins; the reasoning model leans away.

5.3 Causal Evidence: Probe-Direction Steering

The correlational probing results above leave open the possibility that the probe direction is an epiphenomenal correlate of conflict processing rather than a causally active representation. We test this directly by steering Llama-3.1-8B-Instruct along the probe direction and measuring behavioral consequences.

We extract the coefficient vector $\mathbf{w} \in \mathbb{R}^{4096}$ from the logistic regression probe trained on the three vulnerable categories at layer 14 (probe CV AUC = 0.960) and normalize it to $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$. At inference time, we add $\alpha \cdot \hat{\mathbf{w}}$ to the residual stream at L14, sweeping $\alpha \in \{-5, -3, -1, 0, +1, +3, +5\}$ across all 80 vulnerable-category conflict items. Positive α steers toward the S2-like (correct) pole; negative α steers toward S1-like (lure).

Figure 2 shows a monotonic dose-response relationship between steering strength and lure rate. At $\alpha = -5$ (toward S1-like), lure rate rises to 68.8%; at $\alpha = +5$ (toward S2-like), it drops to 31.2%—a total causal swing of 37.6 percentage points. Relative to the unsteered baseline (52.5%), the S2-like direction reduces lure rate by 21.3pp, comparable to Huang et al. [2026]'s 26–32% reduction via contrastive activation addition.

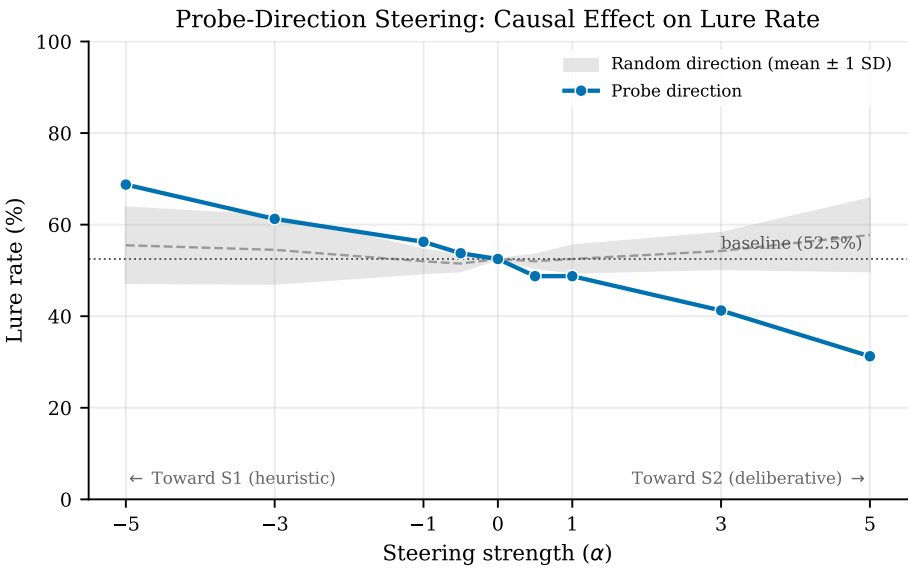

Figure 2: Probe-direction steering on Llama-3.1-8B-Instruct (L14, 80 vulnerable-category conflict items). The probe direction produces a 37.6pp dose-response ($\alpha = -5$: 68.8%; $\alpha = +5$: 31.2%); random directions (gray band) show no systematic effect.

Random-direction control. For each α, we repeat the intervention along five random unit vectors in $\mathbb{R}^{4096}$ and report mean $\pm$ SD. Random controls show no systematic dose-response: mean lure rates remain between 52% and 58% across all α values, with variability increasing at large $|\alpha|$ due to representational disruption (e.g., $57.8\% \pm 7.9\%$ at $\alpha = +5$). The contrast is sharp: the probe direction produces a 37.6pp swing while random directions of identical norm produce none.

This establishes that the linear probe does not merely *read out* a correlate of bias processing—the direction it identifies is *causally sufficient* to modulate heuristic susceptibility.

R1-Distill steering contrast. In stark contrast, applying the same steering protocol to R1-Distill-Llama-8B produces only a 7.5pp swing (10.0% to 2.5%), comparable to random controls. The same representational direction that is causally active in the standard model is *decoupled from behavioral control* in the reasoning model. This dissociation—readable but not writable—provides the strongest evidence that reasoning training reorganizes the causal pathway, not just the representation: the S1-like/S2-like direction remains linearly decodable (AUC $= 0.930$), but perturbing it no longer modulates behavior.

5.4 Within-CoT Probing

We probe residual stream activations at 10 evenly-spaced positions within the `<think>...</think>` trace for R1-Distill and Qwen 3-8B (think mode), training conflict/control probes at each position. If probe AUC increases across the chain, the reasoning process performs genuine computational work on the conflict representation rather than being decorative [Jiang et al., 2025, Lanham et al., 2023]. Full within-CoT trajectories are reported in Appendix A.

5.5 Scale Analysis: 7B vs. 32B

We evaluate OLMo-3.1-32B-Instruct (4.5× more parameters than the 7B variant) to test whether scale mitigates cognitive bias vulnerability. It does not: the overall lure rate *increases* from 14.9% (7B) to 19.6% (32B), with base rate neglect nearly doubling (46% → 74.3%) and framing emerging as a new vulnerability (0% → 30%). Linear probes achieve near-perfect separability (AUC = 0.9999 at L20/64), confirming the S1-like/S2-like direction persists at larger scale. Full per-category breakdowns appear in Appendix B.

5.6 SAE Features and Attention Entropy

SAE analysis of Llama's L19 residual stream identified 41 features with significant conflict/control differential activation (BH-FDR $q < 0.05$), all surviving Ma et al. falsification [Ma et al., 2026]. Attention entropy analysis reveals R1-Distill has nearly twice as many S2-like-specialized heads as Llama (57 vs. 30 of 1024; BH-FDR $q < 0.05$, $|r_{rb}| \geq 0.3$). Details in Appendix C.

6 Discussion

Conflict detection without resolution. The standard model *detects* conflict (AUC = 0.974) yet fails to resolve it (27.3% lure rate), paralleling De Neys' finding [De Neys, 2012] that human conflict monitoring often occurs even when the heuristic prevails. The reasoning model shows *less* distinct encoding (AUC = 0.930) yet dramatically better resolution (2.4%), replicated in OLMo (AUC 0.996 → 0.962; lure 14.9% → 0.9%). Reasoning distillation makes Type 2 processing *autonomous* [Evans, 2019]—internalized so thoroughly that a sharp conflict signal is no longer needed [Venhoff et al., 2025].

Readable but not writable. In Llama, the probe direction is both readable (AUC = 0.974) and writable (37.6pp causal swing); in R1-Distill, it is readable (AUC = 0.930) but not writable (7.5pp, comparable to random controls). This dissociation—reasoning training decouples the representational axis from the behavioral pathway—is direct evidence for the S2-by-default hypothesis [Evans, 2019]. Qwen's think/no-think toggle reveals the converse: thinking reduces lure rates (21% → 7.3%) without changing `P0` encoding (AUC = 0.971 in both modes)—training and inference-time thinking are distinct mechanisms for bias resistance [Ziabari et al., 2025].

Specificity, safeguards, and limitations. Cross-prediction (AUC = 0.378 on immune categories) confirms the probe captures processing mode, not stimulus features; vulnerability is domain- and model-specific [Stanovich et al., 2016]. We apply five controls [Meloux and Barez, 2025, Anonymous, 2026]: Hewitt–Liang selectivity >34pp, cross-prediction falsification, cross-model transfer (AUC > 0.92), random-direction steering baselines (null effect vs. 37.6pp for the probe direction), and Ma et al. SAE falsification (0/41 spurious). Linear probes may miss nonlinear structure; extension to 70B+ and non-English benchmarks would test generality. Multi-seed robustness analysis appears in Appendix D.

Acknowledgments. [Redacted for anonymous review.] Code, benchmark, and activation data will be released upon acceptance.

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A Within-CoT Probing Details

This appendix details the within-chain-of-thought probing protocol for R1-Distill-Llama-8B and Qwen 3-8B (think mode). The analysis tests whether reasoning traces perform genuine computational work on conflict representations or are merely decorative post-hoc rationalizations—a critical question given that only ∼2.3% of reasoning steps causally influence final answers [Jiang et al., 2025, Lanham et al., 2023].

A.1 Method

For each reasoning model–item pair, we extract residual stream activations at five canonical positions within the `<think>...</think>` trace:

- **T0**: First token of the thinking trace (immediately after `<think>`). This captures the "initial impression" before any explicit reasoning.
- **T25**: Token at 25% of the trace length.
- **T50**: Midpoint of the trace.
- **T75**: Token at 75% of the trace length.
- **Tend**: Final token before `</think>`. This captures the representation after reasoning is complete but before the answer is generated.

At each position, we extract the full residual stream (post-MLP, post-residual) at all layers. We then train independent ℓ_2-regularized logistic regression probes (5-fold stratified CV) at each (layer, position) combination on the conflict vs. no-conflict classification, restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms; $n = 160$).

A.2 Expected Analysis

For each model, the primary analysis computes the **AUC trajectory** across the five thinking positions at the peak probing layer identified from the `P0` analysis (L31 for R1-Distill, L34 for Qwen 3-8B think).

Hypothesis 1: Genuine computation. If the reasoning trace performs genuine computational work, we expect probe AUC to *increase* monotonically from T0 to Tend. The magnitude of the T0→Tend AUC gain indicates how much representational sharpening occurs during deliberation. A large gain (e.g., >5pp) would indicate that the model's internal conflict representation is refined through the thinking process.

Hypothesis 2: Performative reasoning. If the model "decides" at T0 and emits decorative reasoning, we expect AUC to be *flat* across positions: the conflict/control distinction is already fully encoded before explicit reasoning begins. Under this hypothesis, T0 AUC ≈ Tend AUC at the peak layer.

Hypothesis 3: Non-monotonic trajectory. A third possibility—observed in some preliminary analyses of other reasoning models—is that AUC *dips* at intermediate positions (T25–T50) before recovering at Tend. This would suggest that intermediate reasoning steps temporarily destabilize representations before converging on a resolution, consistent with the "exploration before exploitation" pattern reported by Boppana et al. [2026].

A.3 Connection to the Qwen Dissociation

The within-CoT analysis directly addresses the Qwen dissociation reported in Section 5.2. At `P0`, Qwen 3-8B produces identical probe signatures in think and no-think modes (AUC = 0.971 at L34 in both). The within-CoT analysis extends this by testing whether the *thinking trace itself* sharpens Qwen's representations. Three outcomes are possible:

1. If Qwen think shows AUC gain from T0 to Tend while R1-Distill does not, the behavioral improvement from thinking is mediated by in-context representational change, not by training-induced encoding differences.
2. If both models show AUC gain, deliberation sharpens representations universally, and the `P0` identity is merely an artifact of the extraction position preceding the thinking trace.
3. If neither model shows AUC gain, the behavioral improvement from thinking operates through a mechanism not captured by linear probes (e.g., nonlinear interactions or attention pattern changes), and the "performative reasoning" hypothesis cannot be rejected.

Note on data availability. Within-CoT probing requires activation extractions at multiple token positions within the reasoning trace. These extractions are computationally intensive (5 positions $\times$ 36 layers $\times$ 470 items) and will be completed upon GPU availability. The protocol, extraction code, and analysis pipeline are finalized; only the compute step remains.

B Scale Analysis: Full Results

We evaluate OLMo-3.1-32B-Instruct (64 layers, 5120 hidden dimensions, 4.5$\times$ the parameters of OLMo-3-7B) on the full 470-item benchmark. This section reports the complete per-category behavioral results and the full 64-layer probe AUC profile.

B.1 Per-Category Behavioral Results: 7B vs. 32B

Table 3 compares per-category lure rates between OLMo-3-7B-Instruct and OLMo-3.1-32B-Instruct.

Key findings. Scale increases *overall* vulnerability (14.9% $\rightarrow$ 19.6%), but the effect is category-specific and non-monotonic:

- **Base rate neglect nearly doubles** (46% $\rightarrow$ 74.3%): the larger model's richer world knowledge may strengthen representativeness-based reasoning, making base rate information harder to integrate.
- **Framing emerges as a new vulnerability** (0% $\rightarrow$ 30%): the 32B model shows sensitivity to gain/loss framing that the 7B model resists entirely. This suggests that scale enables more nuanced sensitivity to surface-level framing cues.
- **Loss aversion and availability are resolved** (33% $\rightarrow$ 0% and 13% $\rightarrow$ 0%): scale provides sufficient capacity to override these heuristics, suggesting they require less specialized processing.
- **Conjunction fallacy is scale-invariant** (50% at both scales), consistent with this being a deeply encoded representational bias rather than a capacity limitation.

Table 3: S1-lure rates (%) on conflict items: OLMo-3-7B-Instruct vs. OLMo-3.1-32B-Instruct (all 11 categories, 235 conflict items). Bold indicates vulnerability (lure rate >10%). "Δ" = 32B − 7B; positive values indicate increased vulnerability at scale.

Category (n conflict)	**OLMo 7B**	**OLMo 32B**	**Δ**	**Direction**
Categories vulnerable at both scales				
Base rate (n=35)	**46.0**	**74.3**	+28.3	↑
Conjunction (n=20)	**50.0**	**50.0**	0.0	—
Categories newly vulnerable at 32B				
Framing (n=20)	0.0	**30.0**	+30.0	↑
Anchoring (n=20)	5.0	**10.0**	+5.0	↑
CRT variants (n=30)	3.0	6.7	+3.7	↑
Categories immune at both scales				
Syllogism (n=25)	0.0	0.0	0.0	—
Arithmetic (n=25)	0.0	0.0	0.0	—
Sunk cost (n=15)	0.0	0.0	0.0	—
Loss aversion (n=15)	**33.0**	0.0	−33.0	↓
Certainty effect (n=15)	0.0	0.0	0.0	—
Availability (n=15)	**13.0**	0.0	−13.0	↓
Overall (11-cat)	**14.9**	**19.6**	+4.7	↑

B.2 Per-Layer Probe AUC (OLMo-3.1-32B-Instruct, 64 Layers)

Table 4 reports the full layer-wise probe AUC for OLMo-3.1-32B-Instruct. Probes are ℓ_2-regularized logistic regression (5-fold stratified CV) trained on conflict vs. no-conflict at the `P0` position (n = 160 items, vulnerable categories).

Table 4: Layer-wise probe AUC for OLMo-3.1-32B-Instruct (64 layers, vulnerable categories, `P0` position, 5-fold CV). Mean ± std over folds. Bold indicates the peak layer. All P2 (control position) probes return AUC = 0.500 exactly.

Layer	**AUC**	**± std**	**Layer**	**AUC**	**± std**
0	0.862	0.010	32	0.997	0.002
1	0.895	0.011	33	0.997	0.002
2	0.905	0.010	34	0.996	0.002
3	0.894	0.013	35	0.996	0.002
4	0.904	0.014	36	0.996	0.002
5	0.908	0.012	37	0.996	0.001
6	0.917	0.011	38	0.997	0.002
7	0.919	0.013	39	0.997	0.002
8	0.919	0.010	40	0.997	0.002
9	0.938	0.006	41	0.996	0.002
10	0.938	0.005	42	0.996	0.002
11	0.946	0.005	43	0.996	0.001

Layer	AUC	± std	Layer	AUC	± std
12	0.975	0.002	44	0.997	0.001
13	0.972	0.002	45	0.997	0.002
14	0.980	0.000	46	0.997	0.001
15	0.987	0.001	47	0.997	0.001
16	0.991	0.001	48	0.997	0.002
17	0.996	0.001	49	0.997	0.002
18	0.997	0.001	50	0.997	0.001
19	0.997	0.001	51	0.997	0.001
20	**0.9999**	**0.0001**	52	0.997	0.002
21	0.9996	0.000	53	0.997	0.001
22	0.9999	0.000	54	0.997	0.001
23	0.999	0.001	55	0.997	0.001
24	0.997	0.001	56	0.997	0.001
25	0.997	0.001	57	0.997	0.001
26	0.997	0.001	58	0.997	0.001
27	0.996	0.001	59	0.997	0.001
28	0.996	0.001	60	0.997	0.001
29	0.997	0.002	61	0.997	0.001
30	0.997	0.002	62	0.997	0.001
31	0.997	0.002	63	0.997	0.001

B.3 Probe AUC Phase Summary

Table 5: Phase-wise probe AUC summary for OLMo-3.1-32B-Instruct. Phases defined as: early (L0–15), mid-early (L16–31), mid-late (L32–47), final (L48–63).

Phase	Layers	Mean AUC	Min AUC	Max AUC
Early	0–15	0.943	0.862	0.987
Mid-early	16–31	0.997	0.991	0.9999
Mid-late	32–47	0.997	0.996	0.997
Final	48–63	0.997	0.997	0.997

Discussion: why scale increases vulnerability. The 32B model achieves near-perfect probe separability (AUC = 0.9999 at L20) yet shows *higher* behavioral vulnerability (19.6% vs. 14.9%). This dissociation—better conflict detection with worse conflict resolution—parallels the Llama vs. R1-Distill pattern at 8B scale and mirrors De Neys' finding in humans that conflict monitoring does not guarantee override [De Neys, 2012].

We propose two complementary explanations for why scale increases vulnerability:

1. **Richer heuristic representations.** Larger models encode more diverse world knowledge, which may strengthen representativeness-based reasoning. A 32B model trained on more data develops stronger priors about category membership (e.g., the "stereotypical lawyer/engineer" association in base rate items), making it harder to override these priors with statistical information.

2. **Expanded surface sensitivity.** The emergence of framing vulnerability at 32B (0% → 30%) suggests that scale enables sensitivity to subtle linguistic cues (gain vs. loss framing) that smaller models simply cannot detect. In this view, scale grants the capacity to be influenced by framing effects that smaller models are too impoverished to represent.

Both explanations are consistent with the "conflict detection without resolution" pattern: the 32B model detects conflict (AUC = 0.9999) but its larger representation space also encodes stronger heuristic attractors that resist override. The plateau in probe AUC from L20 onward (all layers ≥0.996) confirms that scale does not alter the *nature* of the S1-like/S2-like distinction—only its behavioral consequences.

C SAE Features and Attention Entropy

This appendix provides full details on the sparse autoencoder (SAE) feature analysis and attention entropy analysis summarized in Section 5.6.

C.1 SAE Reconstruction Fidelity

We apply a Goodfire SAE (65,536 features) to Llama-3.1-8B-Instruct layer 19 residual stream activations. Table 6 reports reconstruction fidelity on our benchmark items.

Table 6: SAE reconstruction fidelity on Llama-3.1-8B-Instruct L19 activations ($n = 470$ items, `P0` position).

Metric	**Value**
MSE (reconstruction)	0.011
Explained variance (EV)	74.0%
Mean L0 (active features)	47.9 / 65,536
SAE width	65,536

The explained variance of 74% is moderate: the SAE captures the majority of activation variance but not all. Feature-level findings should be interpreted as a lower bound on the true signal, since reconstruction loss may attenuate feature activations.

C.2 Significant Features

After BH-FDR correction ($q < 0.05$) across all 65,536 features, we identify **41 features** with significantly different mean activation between conflict and control items. These features constitute 0.063% of the total SAE dictionary, indicating high specificity.

C.3 Ma et al. Falsification Protocol

All 41 significant features were subjected to the Ma et al. [Ma et al., 2026] falsification protocol, which tests whether a feature's activation is driven by token-level artifacts rather than context-level semantics.

Protocol. For each feature:

1. Identify the top-3 tokens that maximally activate the feature across all benchmark items.

2. Inject each token into 100 random non-cognitive-bias texts (drawn from a held-out corpus of Wikipedia paragraphs, news articles, and code snippets).
3. Measure the feature's activation in the random context.
4. A feature is declared *spurious* if its mean activation in random contexts exceeds 50% of its mean activation in the benchmark context.

Table 7: SAE falsification results (Llama-3.1-8B-Instruct, L19).

Metric	**Value**
Features tested (post-FDR)	41
Spurious (failed falsification)	0
Surviving features	**41**
Spurious rate	0.0%

All 41 features survive falsification. The 0% spurious rate is noteworthy given that Meloux and Barez [2025] report 45–90% spurious rates for purported "reasoning features" in other settings. The low rate here likely reflects our strict initial filtering (BH-FDR $q < 0.05$ with effect-size thresholds) and the fact that cognitive bias processing engages contextual rather than token-level representations.

Trigger token analysis. Inspection of the top-activating tokens for the 41 surviving features reveals that they are predominantly content words related to probability, quantity, and social categorization (e.g., "likely," "percent," "group," "typical") rather than function words or formatting tokens. This is consistent with the features encoding aspects of probabilistic reasoning and category-based judgment that are engaged specifically by the cognitive bias stimuli.

C.4 Attention Entropy: Full Head-Level Analysis

We compare attention entropy patterns between Llama-3.1-8B-Instruct and R1-Distill-Llama-8B at the `P0` position. Attention entropy is computed as the Shannon entropy of each head's attention distribution, normalized by $\log_2(t)$ where t is the sequence length.

Table 8: Summary of head-level attention entropy tests (32 layers $\times$ 32 heads = 1,024 total per model). Significance assessed via rank-biserial correlation (r_{rb}) between conflict and control items, BH-FDR corrected at $q < 0.05$. S2-like-specialized heads additionally require $|r_{rb}| \geq 0.3$.

Metric	**Llama 8B-Inst.**	**R1-Distill**
Total heads tested	1,024	1,024
Significant (BH $q < 0.05$)	473 (46.2%)	515 (50.3%)
Of which conflict-higher	82 (17.3%)	115 (22.3%)
Of which control-higher	391 (82.7%)	400 (77.7%)
S2-like-specialized heads ($q < 0.05$ and $\|r_{rb}\| \geq 0.3$)	30 (2.9%)	57 (5.6%)
Of which conflict-higher	29 (96.7%)	47 (82.5%)
KV-group level	13/256 (5.1%)	24/256 (9.4%)

Key findings.

- R1-Distill has nearly **twice the proportion of S2-like-specialized heads** (5.6% vs. 2.9%) compared to Llama. This gap widens further at the KV-group level (9.4% vs. 5.1%), where GQA non-independence is accounted for.
- The majority of significant heads in both models show *control-higher* entropy (82.7% for Llama, 77.7% for R1-Distill), meaning that conflict items generally elicit more focused (lower entropy) attention patterns. This is consistent with conflict items triggering targeted processing of specific diagnostic elements in the prompt.
- Among S2-like-specialized heads, nearly all show conflict-higher entropy (96.7% for Llama, 82.5% for R1-Distill). These heads attend more broadly on conflict items, potentially integrating information across the full problem statement rather than fixating on heuristic-relevant features.

Layer-wise entropy gap. The entropy gap (conflict − control mean normalized entropy) shows distinct profiles:

- **Llama-3.1-8B:** Peak entropy gap at L7 ($\Delta = -0.016$), concentrated in early-to-mid layers (L4–L14).
- **R1-Distill:** Peak entropy gap at L11 ($\Delta = -0.017$), with a broader distribution extending through L15 and a secondary focus in late layers (L27–L31).

The negative sign indicates that conflict items elicit lower entropy (more focused attention) than control items. The broader distribution in R1-Distill is consistent with reasoning distillation distributing conflict-processing across more layers rather than concentrating it in early attention heads.

D Multi-Seed Robustness

This appendix reports multi-seed evaluation results for Llama-3.1-8B-Instruct and R1-Distill-Llama-8B under sampled decoding (temperature 0.7, top-p = 0.95, 3 seeds: 0, 42, 123). All main-text results use greedy decoding ($T = 0$); this analysis tests whether behavioral findings are robust to the stochasticity introduced by sampling.

D.1 Overall Stability

Table 9: Multi-seed overall lure rates (11 categories, 235 conflict items per seed, sampled decoding at $T = 0.7$). Main-text greedy results are shown for comparison. "σ" = standard deviation across seeds in percentage points.

	Sampled ($T = 0.7$, 3 seeds)				**Greedy**
Model	**Seed 0**	**Seed 42**	**Seed 123**	**Mean ± σ**	**($T = 0$)**
Llama-3.1-8B-Inst.	26.4%	27.2%	28.9%	27.5% ± 1.3pp	27.3%
R1-Distill-Llama-8B	11.5%	11.5%	13.2%	12.1% ± 1.0pp	2.4%*

*The large gap between R1-Distill greedy (2.4%) and sampled (12.1%) reflects the generation strategy dependence discussed in Section D.3.

Llama stability. Llama's overall lure rate is remarkably stable across seeds: 27.5% ± 1.3pp, closely matching the greedy result of 27.3%. This confirms that the behavioral vulnerability is a robust property of the model's representations, not an artifact of a particular decoding path.

R1-Distill: greedy vs. sampled divergence. R1-Distill shows a large gap between greedy (2.4%) and sampled (12.1%) lure rates. Under sampled decoding, the reasoning model's chain-of-thought occasionally takes suboptimal paths that fail to override heuristic lures. The within-seed stability is high (11.5%–13.2%, $\sigma = 1.0$pp), confirming this is a systematic effect of the generation strategy rather than random noise.

D.2 Per-Category Profiles Under Sampling

Table 10 reports per-category lure rates under sampled decoding.

Table 10: Per-category lure rates (%) under sampled decoding ($T = 0.7$, mean ± σ across 3 seeds). "Δ vs. greedy" = sampled mean − greedy. Categories marked with † show instability ($\sigma > 10$pp).

	Llama		**R1-Distill**	
Category	**Sampled**	**Δ greedy**	**Sampled**	**Δ greedy**
Base rate	81.0 ± 3.3	−3.0	1.0 ± 1.6	−3.0
Conjunction	3.3 ± 2.9	−51.7	11.7 ± 2.9	+11.7
Syllogism	41.3 ± 9.2	−10.7	14.7 ± 8.3	+14.7
CRT variants	35.6 ± 5.1	+35.6	35.6 ± 5.1	+32.6
Arithmetic	4.0 ± 0.0	+4.0	5.3 ± 2.3	+5.3
Framing	53.3 ± 2.9	+53.3	31.7 ± 17.6†	+31.7
Anchoring	0.0 ± 0.0	0.0	1.7 ± 2.9	−8.3
Sunk cost	4.4 ± 3.8	+4.4	0.0 ± 0.0	0.0
Loss aversion	0.0 ± 0.0	0.0	22.2 ± 10.2†	+22.2
Certainty eff.	6.7 ± 6.7	+6.7	0.0 ± 0.0	0.0
Availability	8.9 ± 10.2†	−4.1	0.0 ± 0.0	0.0

Category profile shifts. While overall lure rates are stable, category profiles shift substantially under sampled decoding. For Llama, the most dramatic changes are:

- **CRT and framing become highly vulnerable** (0% greedy → 35.6% and 53.3% sampled respectively). These categories are immune under greedy decoding because the maximum-probability token path avoids the lure, but sampling occasionally selects lower-probability heuristic responses.
- **Conjunction drops dramatically** (55% greedy → 3.3% sampled), suggesting that Llama's greedy conjunction errors are a narrow peak in the output distribution that sampling diversifies away from.

For R1-Distill, sampling reveals latent vulnerabilities that greedy decoding masks: CRT (3% → 35.6%), framing (0% → 31.7%), and loss aversion (0% → 22.2%) all emerge as substantially vulnerable when the chain-of-thought can take suboptimal reasoning paths.

Table 11: Generation strategy comparison. Greedy results represent the maximum-probability decoding path; sampled results represent the distributional expectation.

Model	**Greedy**	**Sampled**	$\|\Delta\|$
Llama-3.1-8B-Inst.	27.3%	27.5% ± 1.3pp	0.2pp
R1-Distill-Llama-8B	2.4%	12.1% ± 1.0pp	9.7pp

D.3 Generation Strategy Dependence

Implications. The greedy–sampled divergence is asymmetric: Llama's vulnerability is essentially identical under both strategies ($|\Delta| = 0.2$pp), while R1-Distill's vulnerability is 4× higher under sampling ($|\Delta| = 9.7$pp). This asymmetry has two implications:

1. **Greedy results are maximum-probability, not distributional.** The 2.4% greedy lure rate for R1-Distill reflects the single most-likely decoding path. The 12.1% sampled rate better approximates the model's distributional vulnerability—the expected lure rate over the space of possible reasoning traces.
2. **Reasoning models are more sensitive to generation strategy.** The reasoning model's chain-of-thought introduces a long sequence of stochastic choices, each of which can derail the reasoning process. Standard models generate shorter responses with fewer opportunities for the decoding strategy to influence the outcome.

D.4 Probe Invariance to Generation Strategy

Crucially, probe results at `P0` are **unaffected** by generation strategy, because `P0` activations are deterministic given the prompt. The pre-decision representation is identical regardless of whether the model subsequently generates via greedy or sampled decoding. This confirms that our probing results (AUC values, cross-prediction, steering) reflect stable properties of the model's internal representations rather than artifacts of a particular generation strategy.

The probe–behavior divergence under sampling further supports the "conflict detection without resolution" interpretation: the model *always* encodes the conflict in its `P0` representation (AUC = 0.930 regardless of decoding), but whether it successfully *resolves* that conflict depends on the stochastic reasoning path it takes during generation.